

UNIVERSITY OF MARYLAND, BALTIMORE COUNTY

CMSC 687/487: Introduction to Network Security

Spring 2026

Lectures:	Monday and Wednesday 2:30 PM -3:45 PM
Location:	Janet & Walter Sondheim 409
Instructor:	Dr. Yiwen Hu <huyiwen@umbc.edu >
TA:	Bojing Li <ji18978@umbc.edu
Grader:	TBD
Prerequisite	Undergraduate: CMSC 421 or CMSC 481 (grade of C or better) Graduate: CMSC 341 and CMSC 481

Course Description

This course covers fundamental concepts, principles, and practical techniques for securing computer networks. It examines both traditional and emerging threats, along with defense mechanisms deployed at Internet scale. Major topics include Internet protocol security, web security, mobile malware analysis, network monitoring and anomaly detection, cryptography fundamentals, cloud security, and advanced topics such as 5G and satellite network security. Students will gain hands-on experience with security tools and techniques, develop practical skills through applied exercises, and be inspired to pursue advanced projects and research in network security.

Office Hours

Instructor:	Dr. Yiwen Hu
<i>Office:</i>	ITE 322-A
<i>Office Hours:</i>	Monday, 3:45–4:45 PM (or by appointment)
<i>Email:</i>	huyiwen@umbc.edu

Teaching Assistant:	Bojing Li
<i>Office:</i>	TBD
<i>Office Hour:</i>	TBD
<i>Email:</i>	ji18978@umbc.edu

Textbook and Reading Materials

The textbook below is recommended but not required. Additional reading materials may be shared during lectures to spark your interest and deepen your understanding.

Title: *Computer & Internet Security: A Hands-on Approach*

Authors: Wenliang Du

Edition: Third Edition

Publisher: Wenliang Du (Self Publisher), May 2022

Title: *Network Security Essentials*

Authors: William Stallings

Edition: Six Edition

Publisher: Pearson

Communications

I encourage all students to engage actively in classroom discussions. I value your feedback throughout the semester. You are welcome to share your thoughts during office hours or reach out to me by email.

- All course announcements, materials, and assignment submissions will be handled through **Blackboard**.
- **Blackboard discussion board** is a place for general questions or clarifications. Please do not share solutions or post materials that directly reveal answers.
- You can also **email** us for questions or assistance. When you do, please start your subject line with [CMSC487/687] and send your email to BOTH the instructor and TA using your UMBC email.
 - We'll do our best to reply within 24 hours on weekdays. Emails that don't follow instructions or are sent at the last minute (e.g., right before an assignment is due) may not receive a response, so please plan ahead.

Grading

This section highlights the grading breakdown, grading scale, and important details about exams, homework, projects, and quizzes.

- **Grading Composition:** The course grade will be based on your performance in assignments, projects, quizzes, and exams as follows:
 - Midterm exam: 15%

- Final exam: 20%
- Homework assignments: 21%
- Course Project: 40%
- In-class quizzes (for attendance purposes, no prior announcements): 4%
- **Grading Scale:** Grades are based on the percentage of total points earned (X). Curving is at the discretion of the professor.
 - $90\% \leq X$: A
 - $80\% \leq X < 90\%$: B
 - $70\% \leq X < 80\%$: C
 - $60\% \leq X < 70\%$: D
 - $X < 60\%$: F
- **Exams:** There will be one midterm exam and one final exam. Details regarding exam locations, time slots, and required materials will be announced in class. Makeup exams will not be permitted except in extreme circumstances with valid, documented justification.
- **Homework Assignments:** There will be three homework assignments, may including conceptual questions, exercises, or tasks using network tools. All homework assignments are individual tasks and must be completed by each student independently.
- **Course Project:** Each student will participate in one group project (4 students per group). Detailed requirements for the proposal, video, presentation, and final report (4–6 pages, IEEE/ACM two-column format) will be provided and must be followed. The instructor will suggest high-level project topics, but groups are also welcome to propose their own; all topics must relate to network security. Only proposals approved by the instructor may proceed to subsequent stages. All students are expected to actively participate at every stage, and individual contributions should be clearly indicated in all submissions for grading. The project will be graded as follows:
 - 5% for the project proposal
 - 5% for the midterm video report
 - 10% for the final presentation
 - 20% for the final report and validation of submitted project code/materials
- **Quizzes:** There will be four quizzes during class. Quizzes will not be announced in advance, and no makeup quizzes will be allowed.
- **Deadlines and Late Submission Policy:** The due dates for assignments and the project will be announced on Blackboard, with a deadline of 11:59 PM (UMBC time) on the specified day, unless stated otherwise. Submission instructions will be provided as well. Late assignments and project will be accepted with the following penalties applied to the earned score:

- Up to 12 hours late: **-10%**
- 12–24 hours late: **-20%**
- 24–48 hours late: **-40%**
- More than 48 hours late: **no credit**

Lecture Notes (Tentative)

Lecture	Date	Subject
L01	Jan. 26, 2026	UMBC Class Canceled
L02	Jan. 28, 2026	UMBC Class Canceled
L03	Feb. 2, 2026	Network Security Overview
L04	Feb. 4, 2026	User Authentication (HW1-R)
L05	Feb. 9, 2026	Cryptographic Tools -for confidentiality and integrity
L06	Feb. 11, 2026	Symmetric Encryption and Message Confidentiality (HW1-D)
L07	Feb. 16, 2026	Public Key Cryptography and Message Authentication
L08	Feb. 18, 2026	Public Key Cryptography and Message Authentication (HW2-R)
L09	Feb. 23, 2026	Network Access Control
L10	Feb. 25, 2026	Network Access Control, Transport-Level Security (HW2-Due)
L11	Mar. 2, 2026	Transport-Level Security, IP Security
L12	Mar. 4, 2026	IP Security (Project-Proposal-Due)
L13	Mar. 9, 2026	Exam Review
L14	Mar. 11, 2026	Midterm Exam
	Mar. 16, 2026	Spring Break, no lecture
	Mar. 18, 2026	Spring Break, no lecture
L15	Mar. 23, 2026	Denial-of-service attacks
L16	Mar. 25, 2026	Malicious Software
L17	Mar. 30, 2026	Database Security
L18	Apr. 1, 2026	Intrusion Detection (Mid-Video-Due, 04/03)
L19	Apr. 6, 2026	Firewall and intrusion prevention systems
L20	Apr. 8, 2026	Web and Cloud Security (HW3-R)
L21	Apr. 13, 2026	Web and Cloud Security
L22	Apr. 15, 2026	Wireless Network Security (Wi-Fi) (HW3-Due)
L23	Apr. 20, 2026	Wireless Network Security (IoT)
L24	Apr. 22, 2026	Cellular Network Security (5G)
L25	Apr. 27, 2026	Cellular Network Security (voice, text, data)
L26	Apr. 29, 2026	Cryptocurrencies, Blockchain
L27	May 4, 2026	Project Final Presentation
L28	May 6, 2026	Project Final Presentation, (Project Final Submission 05/08)
L29	May 11, 2026	Final Review
	May 15, 2026	Final Exam (1:00 PM -3:00 PM)

Academic Integrity

This course adheres to the academic honesty policy outlined at <https://academicconduct.umbc.edu/>. The following is a brief summary of the policy:

“Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC’s scholarly community in which everyone’s academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal.”

Please complete your own work and do not seek inappropriate assistance from others. If you are struggling with the course content, it is your responsibility to communicate with the instructor and/or TA so that we can provide assistance.

Homework Assignments: The use of ChatGPT, similar AI tools, or reference code/solutions found online for homework assignments is allowed, provided that you fully disclose how these resources were used. Failure to disclose constitutes an academic integrity violation. Properly disclosed use may receive partial or full credit, depending on the extent of your edits, the achievement of learning outcomes, and the overall impact on the assignment.

Course Projects and Exams: The use of AI tools or AI assistants, including, but not limited to, language models such as ChatGPT, is strictly prohibited for course projects and exams. Any use of AI in these contexts is considered cheating, and there are no exceptions.

UMBC Policies

Accessibility and Disability Accommodations, Guidance and Resources

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at sds.umbc.edu for registration information and office procedures.

SDS email: disAbility@umbc.edu

SDS phone: [410-455-2459](tel:410-455-2459)

If you will be using SDS approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination

[UMBC Policy](#) in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at ecr@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the [Online Reporting/Referral Form](#). Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Mandated Reporters with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Mandated Reporters, per UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#). Faculty and teaching assistants therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their [rights, resources, and support](#). While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator. If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of [Confidential Resources](#) available to support you:

- [Retriever Integrated Health \(Main Campus\)](#): 410-455-2472; Monday - Friday 8:30 a.m. - 5 p.m.; For After-Hours Support, Call 988.
- [Center for Counseling and Well-Being \(Shady Grove Campus\)](#): 301-738-6273; Monday - Thursday 10:00a.m. - 7:00 p.m. and Friday 10:00 a.m. - 2:00 p.m. (virtual) [Online Appointment Request Form](#)

- Pastoral Counseling via [The Gathering Space for Spiritual Well-Being](#): 410-455-6795; i3b@umbc.edu; Monday - Friday 8:00 a.m. - 10:00 p.m.
- [Women, Gender, and Equity Center](#) (open to students of all genders): 410-455-2714; womenscenter@umbc.edu; Monday - Thursday 9:30 a.m. - 5:00 p.m. and Friday 10:00 a.m. - 4 p.m.

Other Resources [Shady Grove Student Resources](#), [Maryland Resources](#), [National Resources](#).

Child Abuse and Neglect Please note that Maryland law and [UMBC policy](#) require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

Pregnant and Parenting Students

UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#) expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's [Office of Equity and Civil Rights](#). Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence – returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting. In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the [Office of Student Disability Services](#).

Religious Observances & Accommodations

UMBC [Policy](#) provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, [UMBC Policy](#) prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the [online reporting/referral form](#) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.